Use Cases and User Stories – ExpenseEase

Project: ExpenseEase – Web/Mobile Expense-Sharing Application **Document type:** Requirements Specification (Use Cases and User Stories) **Course:** Software Engineering – Mid-Semester Project Submission **Version:** 1.0 | **Date:** 15 October 2026

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1. Introduction

ExpenseEase is a web and mobile application that helps groups of people share expenses and keep track of who owes whom. This document describes the system from the point of view of its users. It lists the actors, presents a use-case diagram, gives a detailed description of twelve use cases, and defines eighteen user stories with acceptance criteria. A traceability table at the end connects each user story to its use case.

Scope. The system allows users to register and log in, create groups, add registered users to groups, add expenses using equal, exact-amount and percentage splits, edit and delete expenses, view net balances and simplified debt-settlement suggestions, record settlements, and view an activity feed and dashboard totals. Features outside this scope (for example, receipt scanning, multiple currencies, recurring expenses or payment-gateway integration) are not part of this document.

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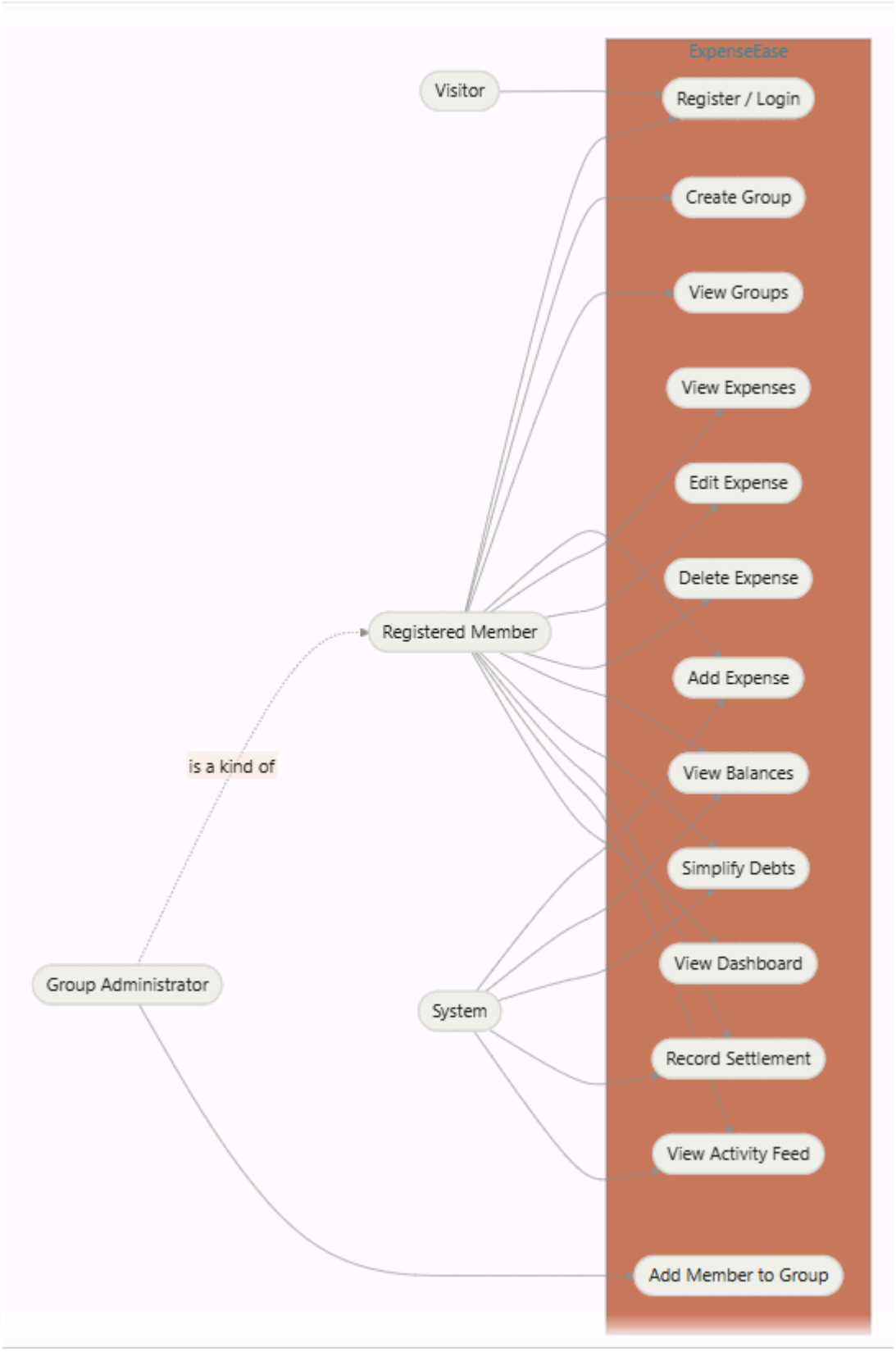
2. Actors and Definitions

Actor	Type	Definition
Visitor	Human (primary)	A person who has opened ExpenseEase but is not logged in. A Visitor can only register or log in.
Registered Member (Member)	Human (primary)	A visitor who has registered and logged in. A Member can create groups, and within groups to which they belong can view information, add expenses, view balances and record settlements. A Member can edit or delete only expenses they created.

Group Administrator (Admin)	Human (primary)	A Member who has administrative rights in a particular group. The creator of a group is its first Admin. An Admin has all Member abilities in that group, and can additionally add registered users to the group and edit or delete any expense in it. Admin rights apply per group only.
System	Secondary (internal)	The ExpenseEase application itself. It validates input, enforces access rules, calculates splits and balances, generates simplified debt suggestions, and writes activity-feed entries.

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3. Use-Case Diagram



Notes on the diagram: The Group Administrator inherits every Member use case and is additionally the only actor who can perform "Add Member to Group". The System participates in use cases where calculation or record-keeping is done automatically. In Section 4, "View Groups" and "View Expenses" are described together as UC-04.

4. Use-Case Descriptions

4.0 Global Business Rules

The following rules apply across the system and are referred to by their IDs in the use-case descriptions.

ID	Business Rule
BR-01	Only authenticated users can access groups and any group data.
BR-02	Only members of a group can view that group's information (members, expenses, balances, settlements and activity).
BR-03	Only a group administrator can add people to a group, and only existing registered users can be added.
BR-04	All money values are stored and calculated as whole numbers of paise (1 rupee = 100 paise). Floating-point arithmetic is not used for money.
BR-05	For every expense, the sum of all participant shares must be exactly equal to the total expense amount.
BR-06	Rounding is deterministic. **Equal split:** base share = floor(total ÷ n); the remainder of r paise (r = total mod n) is given, one paise each, to the first r participants when

	ordered by ascending user ID. **Percentage split:** each share = floor(total × percentage ÷ 100); the leftover paise are given one each to the participants with the largest fractional parts, and ties are broken by ascending user ID. The same input therefore always produces the same shares.
BR-07	An expense can be edited only by its creator or by an administrator of that group.
BR-08	An expense can be deleted only by its creator or by an administrator of that group. A deleted expense is excluded from balance calculations but is retained in the activity history.
BR-09	Debt simplification must preserve every member's final net balance; it may only reduce the number of payments needed. Simplification is a suggestion and does not change stored data.
BR-10	A recorded settlement must update the affected members' balances and must create an activity-feed entry.
BR-11	Every change to group data (group creation, member addition, expense add/edit/delete, settlement) must create an entry in the group's activity feed.
BR-12	The payer and every participant of an expense must be members of the group in which the expense is recorded.

BR-13	Net balance of a member in a group = (total paid by the member for expenses) − (total of the member's shares) + (settlements paid by the member) − (settlements received by the member). A positive value means the member is owed money; a negative value means the member owes money. The net balances of all members of a group always sum to zero.

UC-01 Register / Login

Field	Description
ID and name	UC-01 Register / Login
Primary actor	Visitor
Goal	To create an ExpenseEase account and to log in so that groups and expenses can be accessed.
Preconditions	1. The Visitor has opened the ExpenseEase web or mobile application. 2. For login, the Visitor already has a registered account.
Trigger	The Visitor selects "Register" or "Login".

UC-02 Create Group

Field	Description
ID and name	UC-02 Create Group
Primary actor	Registered Member
Goal	To create a new expense-sharing group and become its administrator.
Preconditions	The Member is logged in.
Trigger	The Member selects "Create Group".
Main success flow	1. The Member selects "Create Group". 2. The System displays a form asking for the group name. 3. The Member enters a name and submits. 4. The System validates the name. 5. The System creates the group, adds the Member as its first member, and assigns the Member the Administrator role for this group. 6. The System records "Group created" in the group's activity feed. 7. The System displays the new group's page.
Alternative / exception flows	**A1 – Empty or over-long name (step 4):** The System rejects the input and asks for a valid name (1 to 60 characters). **E1 – Session expired:** The System redirects to the Login page and the group is not created (BR-01).

Postconditions	Success: A new group exists with exactly one member, who is its Administrator; an activity entry exists. Failure: No group is created.
Business rules	BR-01, BR-11.

UC-03 Add Member to Group

Field	Description
ID and name	UC-03 Add Member to Group
Primary actor	Group Administrator
Goal	To add an existing registered user to a group so that they can share expenses in it.
Preconditions	1. The Administrator is logged in. 2. The Administrator is an administrator of the selected group. 3. The user to be added has a registered account.
Trigger	The Administrator selects "Add Member" on the group page.
Main success flow	1. The Administrator opens the group and selects "Add Member". 2. The System asks for the registered email address of the user to be added. 3. The Administrator enters the email and submits.

	4. The System confirms that the user is registered and is not already a member. 5. The System adds the user to the group with the Member role. 6. The System records "Member added" in the group's activity feed. 7. The System shows the updated member list.
Alternative / exception flows	**A1 – User not registered (step 4):** The System shows "No registered user found with this email". The group is not changed. Inviting unregistered people is outside the system scope. **A2 – Already a member (step 4):** The System informs the Administrator and does not add a duplicate. **E1 – Not an administrator:** If an ordinary Member attempts the action (for example, by direct request), the System denies it with an authorisation error (BR-03). **E2 – Not a group member:** A user who is not in the group cannot open the group at all (BR-02).
Postconditions	Success: The added user is a member of the group and can view it; an activity entry exists. Failure: The member list is unchanged.
Business rules	BR-01, BR-02, BR-03, BR-11.

UC-04 View Groups and Expenses

Field	Description
ID and name	UC-04 View Groups and Expenses

Primary actor	Registered Member
Goal	To see the groups the Member belongs to, and the members and expenses of a selected group.
Preconditions	The Member is logged in.
Trigger	The Member opens "My Groups" or selects a group.
Main success flow	1. The Member opens "My Groups". 2. The System retrieves only the groups of which the Member is a member and displays them. 3. The Member selects a group. 4. The System verifies membership. 5. The System displays the group's member list and its expenses, newest first, showing description, date, total amount, payer and the Member's own share for each expense. 6. The Member may open an expense to see the full split.
Alternative / exception flows	**A1 – No groups (step 2):** The System shows "You are not in any group yet" with a "Create Group" option. **A2 – No expenses (step 5):** The System shows an empty-state message with an "Add Expense" option. **E1 – Not a member (step 4):** If the Member requests a group they do not belong to, the System denies access and reveals no group data (BR-02). **E2 – Not authenticated:** The System redirects to Login (BR-01).
Postconditions	No data is changed. The Member has seen only information they are permitted to see.

Business rules	BR-01, BR-02. Deleted expenses are not shown in the expense list.

UC-05 Add Expense

Field	Description
ID and name	UC-05 Add Expense
Primary actor	Registered Member
Goal	To record a shared expense in a group and divide it among the participants using an equal, exact-amount or percentage split.
Preconditions	1. The Member is logged in. 2. The Member is a member of the group.
Trigger	The Member selects "Add Expense" in a group.

Main success flow	1. The Member selects "Add Expense". 2. The System displays the expense form: description, total amount (in rupees and paise), date, payer (default: the Member), participants, and split type. 3. The Member fills in the details and chooses a split type: Equal, Exact amount, or Percentage. 4. If Exact or Percentage is chosen, the Member enters each participant's amount or percentage. 5. The Member submits the form. 6. The System converts the amount to paise and validates all fields. 7. The System calculates each participant's share: for Equal, using BR-06; for Exact, using the entered amounts; for Percentage, using BR-06. 8. The System checks that the shares sum exactly to the total (BR-05). 9. The System saves the expense with its creator, payer and individual shares. 10. The System updates the group's balances and records "Expense added" in the activity feed. 11. The System displays the expense in the group's expense list.
Alternative / exception flows	**A1 – Exact amounts do not sum to the total (step 8):** The System rejects the expense and displays the difference remaining; nothing is saved. **A2 – Percentages do not sum to 100 (step 8):** The System rejects the expense and displays the current percentage total. **A3 – Invalid amount (step 6):** If the total is zero, negative or has more than two decimal places, the System rejects it. **A4 – No participants selected (step 6):** The System asks for at least one participant.

	A5 – Payer or participant not in the group (step 6): The System rejects the expense (BR-12). **E1 – Not a group member:** The System denies access (BR-02). **E2 – Save failure (step 9):** If the expense cannot be saved, no balance or activity change is made, and the System informs the Member.
Postconditions	Success: The expense is stored with shares that sum exactly to its total; balances are updated; an activity entry exists. Failure: No expense, balance change or activity entry is created.
Business rules	BR-01, BR-02, BR-04, BR-05, BR-06, BR-11, BR-12.

UC-06 Edit Expense

Field	Description
ID and name	UC-06 Edit Expense
Primary actor	Registered Member (as expense creator) or Group Administrator
Goal	To correct the details of an existing expense.
Preconditions	1. The user is logged in and is a member of the group. 2. The expense exists and has not been deleted. 3. The user is the creator of the expense or an administrator of the group.
Trigger	The user selects "Edit" on an expense.

Main success flow	1. The user opens an expense and selects "Edit". 2. The System checks that the user is the creator or a group administrator. 3. The System displays the form filled with the current values. 4. The user changes the required fields (description, amount, date, payer, participants, split type or split values) and submits. 5. The System validates the data and recalculates the shares using the same rules as UC-05, ensuring the shares sum exactly to the new total. 6. The System saves the changes and recalculates the group's balances. 7. The System records "Expense edited" (including who edited it) in the activity feed. 8. The System displays the updated expense.
Alternative / exception flows	**A1 – Shares do not sum to total (step 5):** The System rejects the change; the original expense remains unchanged. **A2 – Invalid input (step 5):** The System shows validation errors and saves nothing. **E1 – Unauthorised user (step 2):** A member who is neither the creator nor an administrator receives an authorisation error and the "Edit" option is not offered (BR-07). **E2 – Expense deleted meanwhile (step 2):** The System informs the user that the expense no longer exists. **E3 – Non-member:** The System denies access (BR-02).
Postconditions	Success: The expense holds the new values with correctly summing shares; balances reflect the change; an activity entry exists. Failure: The expense and balances are unchanged.
Business rules	BR-01, BR-02, BR-04, BR-05, BR-06, BR-07, BR-11, BR-12.

UC-07 Delete Expense

Field	Description
ID and name	UC-07 Delete Expense
Primary actor	Registered Member (as expense creator) or Group Administrator
Goal	To remove an erroneous expense so that it no longer affects balances.
Preconditions	1. The user is logged in and is a member of the group. 2. The expense exists and has not already been deleted. 3. The user is the creator of the expense or an administrator of the group.
Trigger	The user selects "Delete" on an expense.
Main success flow	1. The user opens an expense and selects "Delete". 2. The System checks that the user is authorised to delete it. 3. The System asks the user to confirm the deletion. 4. The user confirms. 5. The System marks the expense as deleted and removes its effect from the group's balances. 6. The System records "Expense deleted" in the activity feed, stating the description, amount and the user who deleted it. 7. The System removes the expense from the expense list and shows a confirmation message.

Alternative / exception flows	**A1 – User cancels (step 4):** Nothing is changed. **E1 – Unauthorised user (step 2):** The System denies the request with an authorisation error (BR-08). **E2 – Already deleted (step 2):** The System informs the user and makes no further change. **E3 – Non-member:** The System denies access (BR-02).
Postconditions	Success: The expense no longer appears in the list or affects balances; the deletion is permanently recorded in the activity history. Failure: The expense and balances are unchanged.
Business rules	BR-01, BR-02, BR-08, BR-11.

UC-08 View Net Balances

Field	Description
ID and name	UC-08 View Net Balances
Primary actor	Registered Member
Goal	To see how much each member of a group is owed or owes.
Preconditions	1. The Member is logged in. 2. The Member is a member of the group.
Trigger	The Member selects "Balances" in a group.

Main success flow	1. The Member selects "Balances". 2. The System verifies membership. 3. The System calculates each member's net balance from all active expenses and recorded settlements (BR-13). 4. The System displays each member's balance in rupees, labelled "gets back" for positive values, "owes" for negative values and "settled up" for zero. 5. The System highlights the Member's own balance.
Alternative / exception flows	**A1 – No expenses (step 3):** All balances are shown as ₹0.00 with the label "settled up". **E1 – Non-member (step 2):** The System denies access (BR-02). **E2 – Not authenticated:** The System redirects to Login (BR-01).
Postconditions	No data is changed. The displayed balances sum to ₹0.00.
Business rules	BR-01, BR-02, BR-04, BR-13.

UC-09 View Simplified Debts

Field	Description
ID and name	UC-01 Register / Login
Primary actor	Visitor
Goal	To create an ExpenseEase account and to log in so that groups and expenses can be accessed.

Preconditions	1. The Visitor has opened the ExpenseEase web or mobile application. 2. For login, the Visitor already has a registered account.
Trigger	The Visitor selects "Register" or "Login".
Main success flow	1. The Member selects "Simplify Debts". 2. The System verifies membership and calculates current net balances (UC-08). 3. The System separates members into those who are owed money (positive balances) and those who owe money (negative balances). 4. The System repeatedly matches the largest debtor with the largest creditor, creating a payment equal to the smaller of the two amounts, until all balances are zero. Ties are resolved by ascending user ID so that the result is deterministic. 5. The System verifies that applying the suggested payments would give every member a final balance of zero, which is equivalent to preserving the original net balances. 6. The System displays the suggested payments in the form "X pays Y ₹amount". 7. The Member may choose a suggestion and proceed to record it (UC-10).
Alternative / exception flows	**A1 – Everyone settled (step 3):** The System shows "All settled up – no payments needed". **E1 – Verification fails (step 5):** The System does not display the suggestions, logs an internal error and shows a general error message. **E2 – Non-member:** The System denies access (BR-02).
Postconditions	No data is changed; the suggestions are not stored as settlements. Net balances remain exactly as before.

Business rules	BR-01, BR-02, BR-09, BR-13.

Example: Net balances are A = +₹300, B = −₹100, C = −₹200. Suggested payments: B pays A ₹100 and C pays A ₹200. Each member's net position is unchanged.

UC-10 Record Settlement

Field	Description
ID and name	UC-10 Record Settlement
Primary actor	Registered Member
Goal	To record that one member has paid money to another member outside the application, so that balances reflect the payment.
Preconditions	1. The Member is logged in. 2. The Member is a member of the group. 3. The Member is either the payer or the receiver in the settlement being recorded. 4. Both the payer and the receiver are members of the group.
Trigger	The Member selects "Record Settlement" or chooses a suggestion from UC-09.
Main success flow	1. The Member selects "Record Settlement". 2. The System displays a form with payer, receiver, amount and date. Values are pre-filled if the Member started from a suggestion. 3. The Member confirms or enters the details and submits. 4. The System validates the data. 5. The System stores the settlement.

	6. The System updates the balances: the payer's balance increases by the amount and the receiver's balance decreases by the same amount. 7. The System records "Settlement recorded" in the group's activity feed. 8. The System displays the updated balances.
Alternative / exception flows	**A1 – Invalid amount (step 4):** If the amount is zero, negative or has more than two decimal places, the System rejects it. **A2 – Payer and receiver are the same (step 4):** The System rejects the settlement. **A3 – Payer or receiver not a group member (step 4):** The System rejects the settlement (BR-12). **E1 – Member not involved (step 4):** A member who is neither payer nor receiver cannot record the settlement. **E2 – Save failure (step 5):** No balance or activity change is made, and the System informs the Member. **E3 – Non-member:** The System denies access (BR-02).
Postconditions	Success: A settlement record exists; balances are updated and still sum to zero; an activity entry exists. Failure: No settlement, balance change or activity entry is created.
Business rules	BR-01, BR-02, BR-04, BR-10, BR-11, BR-12, BR-13.

UC-11 View Activity Feed

Field	Description
ID and name	UC-11 View Activity Feed

Primary actor	Registered Member
Goal	To review a chronological history of what has happened in a group.
Preconditions	1. The Member is logged in. 2. The Member is a member of the group.
Trigger	The Member selects "Activity" in a group.
Main success flow	1. The Member selects "Activity". 2. The System verifies membership. 3. The System retrieves the group's activity entries, newest first. 4. The System displays each entry with the type of action (group created, member added, expense added, expense edited, expense deleted, settlement recorded), the user who performed it, a short description, and the date and time. 5. For long histories the System shows the entries in pages, loading older entries on request.
Alternative / exception flows	**A1 – No activity (step 3):** The System shows an empty-state message. **E1 – Non-member (step 2):** The System denies access and shows no entries (BR-02). **E2 – Not authenticated:** The System redirects to Login (BR-01).
Postconditions	No data is changed. Entries are read-only and cannot be edited or removed by any user.
Business rules	BR-01, BR-02, BR-11.

UC-12 View Dashboard

Field	Description
ID and name	UC-12 View Dashboard
Primary actor	Registered Member
Goal	To see a summary of the Member's financial position across all of their groups.
Preconditions	The Member is logged in.
Trigger	The Member logs in or selects "Dashboard".
Main success flow	1. The Member opens the Dashboard. 2. The System retrieves all groups of which the Member is a member. 3. The System calculates the Member's net balance in each group (BR-13). 4. The System computes the totals: **Total you owe** (sum of the Member's negative group balances), **Total you are owed** (sum of positive group balances) and **Overall net balance** (the difference). 5. The System displays the totals together with the Member's net balance per group and a short list of recent activity.
Alternative / exception flows	**A1 – No groups (step 2):** The System shows totals of ₹0.00 and prompts the Member to create a group. **E1 – Not authenticated:** The System redirects to Login (BR-01).

Postconditions	No data is changed. The totals include only groups of which the Member is currently a member and always agree with the balances shown in UC-08.
Business rules	BR-01, BR-02, BR-04, BR-13.

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5. User Stories

5.1 Summary Table

ID	User Story	Priority
US-01	Register	Must
US-02	Login	Must
US-03	Create Group	Must
US-04	Add Members	Must
US-05	View Groups	Must
US-06	Add Equal-Split Expense	Must
US-07	Add Exact-Share Expense	Must
US-08	Add Percentage-Split Expense	Must

US-09	Deterministic Paise Rounding	Must
US-10	View Expenses	Must
US-11	Edit Own Expense	Should
US-12	Admin Edit Expense	Should
US-13	Delete Expense	Should
US-14	View Net Balances	Must
US-15	View Simplified Debts	Should
US-16	Record Settlement	Must
US-17	View Activity Feed	Should
US-18	View Dashboard Totals	Could

5.2 Detailed User Stories

US-01 Register

Priority: Must

Story: As a visitor, I want to create an account with my name, email and password so that I can use ExpenseEase.

Acceptance criteria

1. **Given** I am a visitor on the registration page, **when** I submit a valid name, a new email address and a password of at least 8 characters, **then** my account is created and I am taken to the Login page.
2. **Given** an account already exists with an email address, **when** I try to register with the same email, **then** the system rejects the registration and shows that the email is already in use.
3. **Given** I am on the registration page, **when** I leave a field empty or enter an invalid email format, **then** the system shows a message for that field and creates no account.
4. **Given** my account has been created, **when** the system stores my password, **then** it is stored in hashed form and is never shown in plain text.

US-02 Login

Priority: Must

Story: As a registered user, I want to log in with my email and password so that I can securely access my groups.

Acceptance criteria

1. **Given** I have a registered account, **when** I enter my correct email and password, **then** I am logged in and the Dashboard is displayed.
2. **Given** I am on the Login page, **when** I enter a wrong email or password, **then** the system shows a general "Invalid email or password" message and does not log me in.
3. **Given** I am not logged in, **when** I try to open a group page directly, **then** I am redirected to the Login page and no group data is shown.

US-03 Create Group

Priority: Must

Story: As a logged-in member, I want to create a group so that I can share expenses with friends, family or colleagues.

Acceptance criteria

1. **Given** I am logged in, **when** I enter a valid group name and submit, **then** a new group is created and I am shown its page.
2. **Given** I have just created a group, **when** I view its member list, **then** I appear as the only member and as the group administrator.
3. **Given** I am creating a group, **when** I submit an empty name or a name longer than 60 characters, **then** the group is not created and an error message is shown.
4. **Given** I have created a group, **when** I open its activity feed, **then** it contains a "Group created" entry.

US-04 Add Members

Priority: Must

Story: As a group administrator, I want to add registered users to my group so that they can take part in shared expenses.

Acceptance criteria

1. **Given** I am an administrator of a group, **when** I enter the email of a registered user who is not yet a member, **then** that user is added to the group and appears in the member list.
2. **Given** I am an administrator, **when** I enter an email that does not belong to any registered user, **then** the system shows that no user was found and the member list is unchanged.
3. **Given** a user is already in the group, **when** I try to add the same user again, **then** the system informs me and does not create a duplicate.
4. **Given** I am an ordinary member and not an administrator, **when** I try to add a person to the group, **then** the system denies the action.
5. **Given** a user has just been added, **when** I open the activity feed, **then** it shows a "Member added" entry.

US-05 View Groups

Priority: Must

Story: As a logged-in member, I want to see a list of my groups so that I can quickly open the one I need.

Acceptance criteria

1. **Given** I belong to one or more groups, **when** I open "My Groups", **then** I see exactly those groups and no others.
2. **Given** I do not belong to any group, **when** I open "My Groups", **then** I see a message saying so, along with an option to create a group.
3. **Given** I am not a member of a particular group, **when** I try to open that group by its address, **then** access is denied and no information about the group is shown.

US-06 Add Equal-Split Expense

Priority: Must

Story: As a group member, I want to add an expense that is divided equally among selected participants so that shared costs are recorded quickly.

Acceptance criteria

1. **Given** I am a member of a group, **when** I add an expense of ₹900.00 paid by me and split equally among three members, **then** each member's share is ₹300.00 and the expense is saved.
2. **Given** I choose the equal split, **when** I select only some group members as participants, **then** only those members receive shares and the other members are not affected.

3. **Given** I am adding an expense, **when** I enter a total of zero, a negative amount, or more than two decimal places, **then** the expense is not saved and an error is shown.
4. **Given** I am adding an expense, **when** I select no participants, **then** the system asks me to select at least one.
5. **Given** the expense is saved, **when** I view group balances, **then** the payer's balance has increased and each participant's balance has decreased by their share.

US-07 Add Exact-Share Expense

Priority: Must

Story: As a group member, I want to enter the exact amount each participant owes so that unequal costs can be recorded accurately.

Acceptance criteria

1. **Given** I am adding an expense of ₹500.00, **when** I choose the exact-amount split and enter ₹200.00, ₹150.00 and ₹150.00 for three participants, **then** the expense is saved with those exact shares.
2. **Given** I choose the exact-amount split, **when** the entered amounts add up to ₹480.00 for a total of ₹500.00, **then** the expense is not saved and the system shows that ₹20.00 remains unassigned.
3. **Given** I choose the exact-amount split, **when** the entered amounts add up to more than the total, **then** the expense is not saved and the system shows the excess amount.
4. **Given** I choose the exact-amount split, **when** I enter a negative amount for a participant, **then** the system rejects the entry.

US-08 Add Percentage-Split Expense

Priority: Must

Story: As a group member, I want to divide an expense by percentages so that each person pays a proportionate share.

Acceptance criteria

1. **Given** I am adding an expense of ₹1,000.00, **when** I choose the percentage split with 50%, 30% and 20%, **then** the shares saved are ₹500.00, ₹300.00 and ₹200.00.
2. **Given** I choose the percentage split, **when** the percentages add up to 95% or 105%, **then** the expense is not saved and the system shows the current percentage total.
3. **Given** I choose the percentage split, **when** I enter a negative percentage or one with more than two decimal places, **then** the system rejects the entry.
4. **Given** the percentage split produces fractional paise, **when** the expense is saved, **then** the shares are rounded according to the deterministic paise rounding rule and sum exactly to the total.

US-09 Deterministic Paise Rounding

Priority: Must

Story: As a group member, I want every split to be calculated to the exact paisa with a fixed rounding rule so that the shares always add up to the expense total and the result never varies.

Acceptance criteria

1. **Given** an expense of ₹100.00 (10,000 paise) is split equally among three participants, **when** the shares are calculated, **then** they are ₹33.34, ₹33.33 and ₹33.33, with the extra paisa given to the participant with the lowest user ID, and the sum is exactly ₹100.00.
2. **Given** an expense of ₹10.01 (1,001 paise) is split by 50%, 30% and 20%, **when** the shares are calculated, **then** they are ₹5.01, ₹3.00 and ₹2.00, because the leftover paisa goes to the participant with the largest fractional part, and the sum is exactly ₹10.01.
3. **Given** any equal or percentage split, **when** the same expense data is calculated more than once, **then** the same shares are produced every time.
4. **Given** any expense is saved or edited, **when** the shares are stored, **then** their sum equals the total amount exactly, with no difference of even one paisa.
5. **Given** the system performs any money calculation, **when** values are stored, **then** they are held as whole paise and not as floating-point numbers.

US-10 View Expenses

Priority: Must

Story: As a group member, I want to view all expenses of my group so that I can see who paid for what.

Acceptance criteria

1. **Given** I am a member of a group with recorded expenses, **when** I open the group, **then** I see the expenses newest first with description, date, total, payer and my share.
2. **Given** I open one expense, **when** its details are displayed, **then** I can see every participant's share and the sum equals the total.
3. **Given** an expense has been deleted, **when** I view the expense list, **then** that expense is not shown.
4. **Given** I am not a member of the group, **when** I try to view its expenses, **then** access is denied.

US-11 Edit Own Expense

Priority: Should

Story: As an expense creator, I want to edit an expense I added so that I can correct mistakes in the amount, payer, participants or split.

Acceptance criteria

1. **Given** I created an expense, **when** I change its total or split values and submit valid data, **then** the expense is updated and the shares are recalculated to sum exactly to the new total.
2. **Given** I am editing my expense, **when** the new shares do not sum to the new total, **then** the change is rejected and the original expense remains unchanged.
3. **Given** I have edited my expense, **when** I view group balances, **then** they reflect the updated expense only and not the old values.
4. **Given** I have edited my expense, **when** I open the activity feed, **then** an "Expense edited" entry names me as the editor.
5. **Given** another member created an expense and I am not an administrator, **when** I look at that expense, **then** I cannot edit it.

US-12 Admin Edit Expense

Priority: Should

Story: As a group administrator, I want to edit any expense in my group so that errors made by others can be corrected.

Acceptance criteria

1. **Given** I am an administrator of a group, **when** I edit an expense created by another member and submit valid data, **then** the expense is updated and balances are recalculated.
2. **Given** I have edited another member's expense, **when** I open the activity feed, **then** the "Expense edited" entry names me as the editor and not the original creator.
3. **Given** I am an administrator only of Group A, **when** I try to edit an expense in Group B where I am an ordinary member, **then** the system denies the edit unless I created that expense.
4. **Given** I am an administrator, **when** my edit would make the shares differ from the total, **then** the system rejects the change.

US-13 Delete Expense

Priority: Should

Story: As an expense creator or group administrator, I want to delete an erroneous expense so that it no longer affects balances.

Acceptance criteria

1. **Given** I created an expense or I am the group administrator, **when** I select "Delete" and confirm, **then** the expense is removed from the expense list.
2. **Given** an expense has been deleted, **when** I view group balances, **then** the balances no longer include that expense.
3. **Given** I am an ordinary member who did not create the expense, **when** I try to delete it, **then** the system denies the action.
4. **Given** I select "Delete", **when** I cancel the confirmation, **then** the expense remains unchanged.

5. **Given** an expense has been deleted, **when** I open the activity feed, **then** it shows an "Expense deleted" entry with the description, amount and the user who deleted it.

US-14 View Net Balances

Priority: Must

Story: As a group member, I want to see each member's net balance so that I know who owes money and who is owed.

Acceptance criteria

1. **Given** I am a member of a group with expenses, **when** I open "Balances", **then** each member's balance is shown as "gets back", "owes" or "settled up" with the amount in rupees.
2. **Given** the balances are displayed, **when** I add them all together, **then** the total is exactly ₹0.00.
3. **Given** a group has no expenses, **when** I open "Balances", **then** every member is shown as settled up with ₹0.00.
4. **Given** an expense is added, edited or deleted, or a settlement is recorded, **when** I reopen "Balances", **then** the figures reflect that change.
5. **Given** I am not a member of the group, **when** I request its balances, **then** access is denied.

US-15 View Simplified Debts

Priority: Should

Story: As a group member, I want to see a simplified list of payments that settles all debts so that the group needs as few transactions as possible.

Acceptance criteria

1. **Given** net balances of A = +₹300.00, B = −₹100.00 and C = −₹200.00, **when** I open "Simplify Debts", **then** the system suggests that B pays A ₹100.00 and C pays A ₹200.00.
2. **Given** simplified payments are displayed, **when** all suggested payments are applied, **then** every member's final balance is zero, showing that the original net balances were preserved.
3. **Given** all members are already settled, **when** I open "Simplify Debts", **then** the system shows that no payments are needed.
4. **Given** I only view the suggestions, **when** I leave the page, **then** no balances, settlements or activity entries have been changed.
5. **Given** the same balances, **when** suggestions are generated more than once, **then** the same list is produced each time.

US-16 Record Settlement

Priority: Must

Story: As a group member, I want to record a payment made to another member so that our balances are updated.

Acceptance criteria

1. **Given** B owes A ₹100.00, **when** B or A records a settlement of ₹100.00 from B to A, **then** the settlement is saved and B's and A's balances are adjusted accordingly.
2. **Given** a settlement has been recorded, **when** I open the activity feed, **then** it contains a "Settlement recorded" entry showing payer, receiver and amount.
3. **Given** I am recording a settlement, **when** I enter an amount of zero or less, or choose the same person as payer and receiver, **then** the settlement is rejected.
4. **Given** I am recording a settlement, **when** the payer or receiver is not a member of the group, **then** the settlement is rejected.
5. **Given** a settlement has been recorded, **when** I view the net balances, **then** the balances still add up to ₹0.00.
6. **Given** I am a member who is neither the payer nor the receiver, **when** I try to record that settlement, **then** the system does not allow it.

US-17 View Activity Feed

Priority: Should

Story: As a group member, I want to see a history of group activity so that I can understand how balances have changed.

Acceptance criteria

1. **Given** I am a member of a group, **when** I open "Activity", **then** I see entries newest first, each showing the action, the user who performed it, and the date and time.
2. **Given** an expense was added, edited or deleted, a member was added, or a settlement was recorded, **when** I view the feed, **then** a corresponding entry exists for each of these actions.
3. **Given** I am viewing the feed, **when** I try to change or remove an entry, **then** the system provides no way to do so.
4. **Given** I am not a member of the group, **when** I request its activity feed, **then** access is denied.
5. **Given** the group has no activity beyond its creation, **when** I open "Activity", **then** I see only the "Group created" entry.

US-18 View Dashboard Totals

Priority: Could

Story: As a logged-in member, I want a dashboard showing my total balances across all groups so that I can see my overall position at a glance.

Acceptance criteria

1. **Given** I belong to several groups, **when** I open the Dashboard, **then** it shows "Total you owe", "Total you are owed" and "Overall net balance" across all of my groups.

2. **Given** I owe ₹250.00 in Group A and am owed ₹400.00 in Group B, **when** I open the Dashboard, **then** it shows ₹250.00 owed by me, ₹400.00 owed to me and an overall net balance of +₹150.00.
3. **Given** I belong to no groups, **when** I open the Dashboard, **then** all totals show ₹0.00 and I am prompted to create a group.
4. **Given** I have recorded a settlement or changed an expense, **when** I reopen the Dashboard, **then** the totals have been updated to match the group balances.
5. **Given** other groups exist in which I am not a member, **when** the Dashboard totals are calculated, **then** only groups of which I am currently a member are included.

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6. Traceability Table

User Story	Title	Use Case(s)	Priority
US-01	Register	UC-01 Register / Login	Must
US-02	Login	UC-01 Register / Login	Must
US-03	Create Group	UC-02 Create Group	Must
US-04	Add Members	UC-03 Add Member to Group	Must
US-05	View Groups	UC-04 View Groups and Expenses	Must
US-06	Add Equal-Split Expense	UC-05 Add Expense	Must
US-07	Add Exact-Share Expense	UC-05 Add Expense	Must
US-08	Add Percentage-Split Expense	UC-05 Add Expense	Must

US-09	Deterministic Paise Rounding	UC-05 Add Expense, UC-06 Edit Expense	Must
US-10	View Expenses	UC-04 View Groups and Expenses	Must
US-11	Edit Own Expense	UC-06 Edit Expense	Should
US-12	Admin Edit Expense	UC-06 Edit Expense	Should
US-13	Delete Expense	UC-07 Delete Expense	Should
US-14	View Net Balances	UC-08 View Net Balances	Must
US-15	View Simplified Debts	UC-09 View Simplified Debts	Should
US-16	Record Settlement	UC-10 Record Settlement	Must
US-17	View Activity Feed	UC-11 View Activity Feed	Should
US-18	View Dashboard Totals	UC-12 View Dashboard	Could

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